

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		$K_c = \frac{[CrO_4^{2-}]^2 [H^+]^2}{[Cr_2O_7^{2-}] [H_2O]}$		1		1		
		(ii)		1 dm ³ contains 1000g of water (1) $\frac{1000}{18.02} = 55.5$ in 1 dm ³ (1)	1	1		2		
		(iii)		$[H^+]^2 = \frac{K_c \times [Cr_2O_7^{2-}] \times [H_2O]}{[CrO_4^{2-}]^2}$ (1) $[H^+]^2 = 2.356 \times 10^{-12}$ or $[H^+] = 1.535 \times 10^{-6}$ mol dm ⁻³ (1) pH = 5.81 (1) student is incorrect as this solution is acidic (1)		1	1 1 1	4	3	
		(iv)		if K_c increases the reaction is shifting to the right (1) if a reaction shifts to right when heated it must be endothermic (1)		1	1	2		
				Question 13 total	1	4	8	13	3	0